Tutorial 08

Characteristic Function, Markov’s and Chebyshev’s Inequalities, Law of Large Numbers

Xujie Zhao

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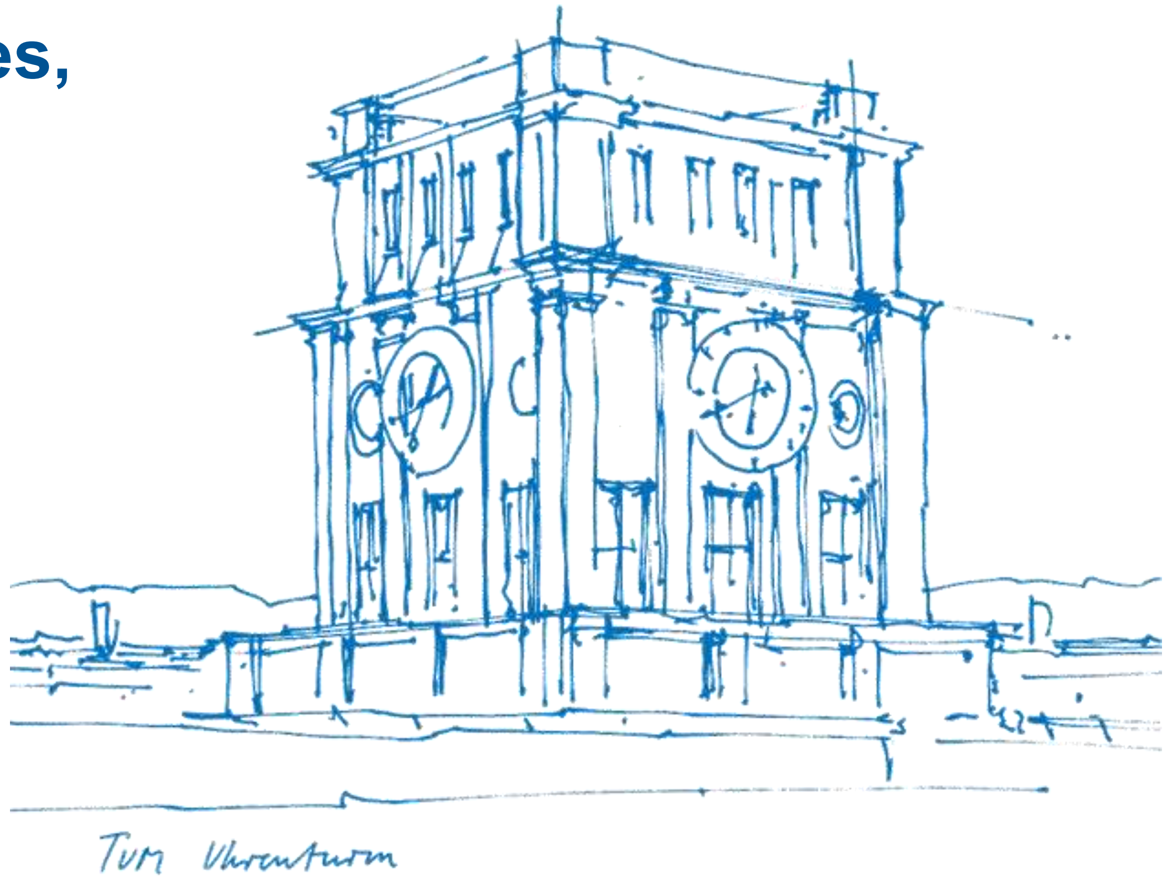

About Tutorial

- **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
  - 14:15-15:45 Tue, 00.08.053 (B42)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

- **Contact:** xujie.zhao@tum.de

➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercises (with solution)

I. Recap

- **Characteristic Function**
  - Definition
  - Property
- **Markov’s and Chebyshev’s Inequalities**
  - Definition
- **(Weak and Strong) Law of Large Numbers**
  - Definition

II. Exercise Sheet

Exercise 1 (sum of independent Poissons is Poisson). Let $X \sim \mathrm{Poi}(\lambda_1)$ and $Y \sim \mathrm{Poi}(\lambda_2)$ be independent RVs. Show that $X + Y \sim \mathrm{Poi}(\lambda_1 + \lambda_2)$.
[Hint: Compute the characteristic function of a Poisson RV and use properties of characteristic functions.]

II. Exercise Sheet

Exercise 2 (server overload). A web server receives requests from two types of users: Type A and Type B. Let N_A be the number of requests from Type A users in an hour, with $\mathbb{E}[N_A] = 50$ and $\text{var}[N_A] = 50$. Similarly, N_B is the number of requests from Type B users in an hour, with $\mathbb{E}[N_B] = 100$ and $\text{var}[N_B] = 100$. Assume N_A and N_B are independent. The server has a capacity to handle $C = 200$ requests per hour. If the total number of requests $N_T = N_A + N_B$ exceeds C, the server is overloaded and breaks down.

(i) Calculate the expected total number of requests per hour, $\mathbb{E}[N_T]$, and the variance of the total number of requests per hour, $\text{var}[N_T]$.

(ii) Using Chebyshev's inequality, provide an upper bound on the probability that the server is overloaded, i.e., on $P(N_T > C)$.

(iii) Assuming N_T is non-negative (quite plausibly), use Markov's inequality and $\mathbb{E}[N_T]$ to provide another upper bound on $P(N_T > C)$. Compare it to the bound in (ii) and explain the difference.

(iv) What would be an adequate distribution to model N_A, N_B and why?

(v) One can show that for $X \sim \text{Poi}(\lambda_1)$, $Y \sim \text{Poi}(\lambda_2)$ and $X \perp\!\!\!\perp Y$, we have that $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$. Using this result, calculate the probability of server overload $P(N_T > C)$ under this model. (You can use a computer for this. For example, `scipy.stats` has functions for the cdfs of most common distributions.) Compare this value with the bounds obtained in (ii) and (iii). Discuss the differences and the utility of the bounds versus an exact calculation for a specific distribution.

II. Exercise Sheet

Exercise 3. Let $(\Omega, \mathcal{A}, P)$ be a probability space, and (X, Y) a $\mathbb{R}^2$-valued RV that is uniformly distributed on the disc

$$\overline{D^2} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\} .$$

The density of this uniform distribution is constant on the disc and has value π, since the area of the unit disc is π. That is,

$$p_{X,Y}(x, y) = \frac{1}{\pi} \chi_{\overline{D^2}}(x, y) .$$

(i) Compute the marginal densities $p_X(x)$ and $p_Y(y)$.

(ii) Compute the means and variances $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathrm{var}[X]$, $\mathrm{var}[Y]$.
[Hint: For $\mathbb{E}[X^2]$, consider the substitution $x = \sin(u)$ and look up trigonometric identities for $\sin(2u)$ as well as for expressing $\sin(x)^2$ in terms of a cos. Use the symmetry between X and Y.]

(iii) Check whether $X \perp\!\!\!\perp Y$ and whether X and Y are uncorrelated.
[Hint: Use polar coordinates to perform the integral for the covariance. Remember that you have to transform the integration boundaries and add the Jacobian determinant in the integral.]

II. Exercise Sheet

Exercise 4 (Optional (coding): Monte Carlo (MC) integration). The laws of large numbers (we will see those formally in the lecture) imply that we can *estimate* the mean $\mathbb{E}[Y]$ of an RVRV Y by "averaging n iid samples," that is

$$\mu := \mathbb{E}[Y] \approx \frac{1}{n}\sum_{i=1}^{n} Y_i \,.$$

This can be used for numerical "quadrature," that is, for estimating integrals $I := \int_U f(x)\,dx$ over some bounded domain $U \subsetneq \mathbb{R}^d$. By generating/sampling samples $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{Unif}(U)$ (that is, all X_i are iid uniform), the estimate using the actual instantiations $x_1, \ldots, x_n$ reads

$$\hat{I}_n := \frac{1}{n}\sum_{i=1}^{n} f(x_i) \approx \frac{1}{\lambda^d(U)} \int_U f(x)\,dx \,.$$

Use this concept to numerically approximate the integral of $f(x) := 10 - x_1^2 - x_2^2$ over the unit disc

$$D^2 = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 < 1\}$$

by implementing the following steps:

(i) Draw n samples uniformly independently inside D^2. (Make sure they are in D^2 and not $(-1, 1)^2$.)

(ii) Estimate the measure of D^2 with these samples in a suitable way, that is, can you use these samples to estimate the area of $D^2 = \pi$.

(iii) Then use the samples to approximate the integral $\int_{D^2} f(x)\,dx$. Compare this to the true analytical value of the integral.

III. Additional Exersices

Exercise 5. The lecture Discrete Probability Theory is attended by 1108 students. Based on experience, the teaching staff knows that, on average, only two out of three students take the final exam. Since the teaching staff is environmentally conscious, they would like to print as few exam sheets as possible. It is assumed that the students make their decisions about whether to take the exam independently of one another. Determine the minimum number of exam sheets that the teaching staff must provide so that these are insufficient with probability at most 10%.

Hint (Chernoff-bound for sum of 0-1-RVs): For $X_1, X_2, \cdots X_n$, they are independent from each other and Bernoulli distributed with $P(X_i=1)=p_i$. Let $X=\sum_{i=1}^{n} X_i$, $\mu=E(X)=\sum_{i=1}^{n} p_i$, then for every $\delta>0$, it holds $P(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$; for $0<\delta\leq 1$, $P(X \geq (1+\delta)\mu) \leq e^{-\mu\delta^2/3}$

For every student X_i, $X_i=0$ for not take the exam; $X_i=1$ for take the exam with $P(X_i=1)=p_i$.

Let $X=\sum_{i=1}^{n} X_i$, $\mu=E(X)=\sum_{i=1}^{n} p_i = 1108 \times \frac{2}{3} = 738.7$.

With Chernoff-bound formula, $(1+\delta)\mu$ is the number of exams to be prepared and $e^{-\mu\delta^2/3}=10\%=0.1$.

Solve that $-\mu\delta^2/3 = \ln 0.1 = -\ln 10$, $\delta=\sqrt{\frac{3\ln 10}{738.7}}=0.0967$.

So $(1+\delta)\mu = 1.0967 \times 738.7 \approx 810$, at least 810 sheets needed.

Materials